Hopfield Networks

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1 Introduction

1.1 Motivation

The Hopfield network [6] is a model for associative memory in the human brain. What is associative memory and why is it important? Consider the following sentence with missing letters: c*n y*u r**d *h*s. Odds are you were able to read it because your brain contains memories - patterns reinforced by repeated stimulus - strongly associated with that noisy sentence. This enabled you to fill in the missing information, clear the noise, and construct a complete sentence. As the world is noisy, associative memory is an important tool that enables brains to generalize. A brain that sees an orange with dirt on its exocarp benefits from associating it with oranges seen before. Building synthetic models of associative memory allows us to interact with the world via technology more robustly.

1.2 Code Review

The original paper did not have any accompanying code, and we do not know which programming language Hopfield used in 1982. Given the early date, the language was probably either FORTRAN or C. We directly implemented Hopfield networks and dense associative memories in Python and JavaScript. The relevant Python code is in the attached `Final Project Hopfield.ipynb` file.

1.3 Outline

In his 1982 paper, Hopfield recognizes two limitations of his model: the $0.15N$ capacity limit and the inability to distinguish between similar patterns. He also suggests a generalization of the network to asymmetric weights. Our paper aims to look deeper into those two limitations as well as the generalization. Section 2 starts with a recap of the mathematics behind the original Hopfield Model and moves on to explain the capacity limit by analogy to the Ising model. Section 3 addresses the problem of being unable to store similar patterns and discusses dense associative memories, a generalization of the original model proposed by Krotov and Hopfield [8] to deal with this problem (as well as the capacity limit). Section 4 looks at biological implications of the model, including an application to an LIF neuron network and a discussion of asymmetric weights and their applications in biology.

2 The Classical Model

2.1 Mathematical Description

Consider a system of N binary neurons, each of which could be in two possible states: firing (represented by $+1$) or non-firing (represented by -1). A *pattern* stored by the binary neurons is a binary string of length N, where the ith position in the string is ± 1 depending on whether the ith neuron is firing or not when the network is remembering this pattern. For example, a 3-neuron pattern $(1, -1, -1)$ means that the first neuron is firing while the second and third are not. Let M be the number of patterns we are storing. We denote a single pattern by

$$\boldsymbol{\xi}^{\mu}, \quad \mu = 1, \cdots, M.$$

If we want to refer to the ith component of the pattern $\boldsymbol{\xi}^{\mu}$, we use the notation ξ_i^{μ}. As a rule, vectors are written in bold script (like $\boldsymbol{\xi}^{\mu}$), and components of vectors or matrices are written in normal math script (like ξ_i^{μ}).

The classical model is best understood as an energy minimization problem. Given a starting pattern $\boldsymbol{\sigma}$, we successively update the pattern so as to minimize an energy function. Ideally, when we reach a minimum of the energy function, the resulting pattern should correspond exactly to one of the stored patterns. Since we are *minimizing* the energy, we want *high* overlap between the trial and stored patterns to correspond to *low* energy. Overlap is conveniently computed using the dot product, so we might consider the energy function[1]

$$E(\boldsymbol{\sigma}) = -\frac{1}{2N} \sum_{\mu=1}^{M} (\boldsymbol{\xi}^{\mu} \cdot \boldsymbol{\sigma})^2. \tag{1}$$

This is simply a sum over the squared dot products between the trial state and the stored states. But note that this formula could be conveniently rewritten as a quadratic form:

$$E(\boldsymbol{\sigma}) = -\frac{1}{2N} \sum_{\mu=1}^{M} (\boldsymbol{\xi}^{\mu} \cdot \boldsymbol{\sigma})^2 = -\frac{1}{2N} \sum_{\mu=1}^{M} (\boldsymbol{\sigma} \cdot \boldsymbol{\xi}^{\mu})(\boldsymbol{\xi}^{\mu} \cdot \boldsymbol{\sigma}) = -\frac{1}{2N} \sum_{\mu=1}^{M} \boldsymbol{\sigma}^T \boldsymbol{\xi}^{\mu} (\boldsymbol{\xi}^{\mu})^T \boldsymbol{\sigma} := -\frac{1}{2} \boldsymbol{\sigma}^T \mathbf{W} \boldsymbol{\sigma},$$

where we have defined the **Hopfield weight matrix**

$$\mathbf{W} := \frac{1}{N} \sum_{\mu=1}^{M} \boldsymbol{\xi}^{\mu} (\boldsymbol{\xi}^{\mu})^T. \tag{2}$$

Now suppose that we flip the ith component of $\boldsymbol{\sigma}$ from σ_i to $-\sigma_i = \sigma_i + \Delta\sigma_i$. Then the energy change caused by this flip is[2]

$$\Delta E = -\Delta\sigma_i \sum_{j \neq i} W_{ij} \sigma_j. \tag{3}$$

[1] A simple linear sum over all the dot products would not work. As Hopfield pointed out in his original paper, if we choose the energy function to be linear, we get something akin to a "linear associative net", which stores input-output pairings $S_i \to O_i$. When a linear associative net is presented with a noisy input $\sum_i S_i$, it generates the meaningless output $\sum_i O_i$. The nonlinearity provided by the quadratic energy function enables the network to converge to a minimum corresponding to a single pattern.

[2] For a rigorous proof, see Section 4.2.

Thus, if we want ΔE to be *negative*, we want $\Delta\sigma_i$ to always have the same sign as $\sum_{j\neq i} W_{ij}\sigma_j$, resulting in the update rule

$$\sigma_i^{(t+1)} = \operatorname{sgn}\left(\sum_{j\neq i} W_{ij}\sigma_j^{(t)}\right), \tag{4}$$

where $\boldsymbol{\sigma}^{(t)}$ is the state at time step t. Since it is inconvenient to carry around the restriction $j \neq i$ in the summation, it is conventional to modify the definition (2) for the weight matrix:

$$\mathbf{W} := \frac{1}{N}\sum_{\mu=1}^{M} \boldsymbol{\xi}^\mu (\boldsymbol{\xi}^\mu)^T, \quad W_{ii} = 0. \tag{5}$$

Note that this does not affect the interpretation of the energy as the sum of squares of dot products, since the diagonal terms always contribute a fixed amount to the energy,[3] and since we care about the derivative of the energy and not its actual value, this constant could be subtracted without difficulty. Note that this weight matrix is strongly illustrative of Hebbian learning. If we add a new pattern $\boldsymbol{\xi}^{M+1}$ to our current ensemble of patterns, the matrix elements are updated as

$$\Delta W_{ij} = \xi_i^{M+1}\xi_j^{M+1},$$

so $\Delta W_{ij} = 1$ if neurons i and j exhibit the same behavior on the pattern $\boldsymbol{\xi}^{M+1}$ and $\Delta W_{ij} = -1$ if one neuron fires while the other does not.

2.2 Storage Capacity and Phase Transitions

2.2.1 The Ising analogy: insights from statistical physics

Roughly, let the storage capacity of the Hopfield network be the largest number of patterns M such that the network can reliably retrieve any of its memories reliably. That is, if any of the $\boldsymbol{\xi}^\mu$ is noised and then fed into the network, it will converge to $\boldsymbol{\xi}^\mu$ with probability at least $1-\epsilon$.

We can first ask why the Hopfield network has a well-defined storage capacity at all. One might naïvely have expected the retrieval accuracy of the network to degrade smoothly as M increases for fixed N. As you try to load more memories into a model of fixed size, the average overlap between memories increases, giving rise to fused minima and spurious attractor states that the network can converge to (more on those in 3.1). It's not *prima facie* clear that there should be any M at which retension accuracy falls sharply. But there does turn out to be such a threshold, and the reason why can be illuminated by analogizing the Hopfield net to the Ising model.

The Ising model[4] comes from condensed matter physics, where it is used to model the behavior of a ferromagnet. The macroscopic magnet is modeled as an array of N magnetic dipoles ("spins") that may each be in one of two orientations. We represent the ith spin being in the up orientation by $s_i = 1$ and the ith spin being in the down orientation by $s_i = -1$. The energy of an array in the state $\boldsymbol{s}$ is then given by

$$E(\boldsymbol{s}) = -\frac{1}{2}\sum s_j J_{ij} s_i = -\frac{1}{2}\boldsymbol{s}^T J \boldsymbol{s}, \tag{6}$$

[3] Note that the diagonal contribution to the energy is $\sum_i W_{ii}\sigma_i^2 = \sum_i W_{ii} = M$, independent of $\boldsymbol{\xi}^\mu$ or $\boldsymbol{\sigma}$.

[4] For more on the Ising model, we recommend §8.2 in *An Introduction to Thermal Physics* by Daniel Schroeder [15].

where J is a matrix whose entries represent the physical distances between spins. J is non-negative because it is more energetically favorable for two spins to align than for them to oppose, J is symmetric because the distance from spin i to spin j is the same as the distance from spin j to spin i, and J's diagonal entries are zero because we assume there is no external applied magnetic field. All three of these properties should remind us of the Hopfield weight matrix. Indeed, comparing equations 1 and 6 reveals that the Ising magnet and the Hopfield net have the same energy function.

If an Ising magnet is held in thermal equilibrium with a reservoir at temperature T, the Boltzmann distribution tells us the probability of finding the system in state $\boldsymbol{s}$ will be

$$P(\boldsymbol{s}) = \frac{\exp\big(-E(\boldsymbol{s})/k_{\mathrm{B}}T\big)}{Z}, \tag{7}$$

where Z is a normalization constant sometimes known as the partition function. This should make us wonder what quantity might play the role for the Hopfield network that temperature plays for the Ising model. Here is a heuristic demonstration that M/N plays this role for the Hopfield network. Consider a starting pattern $\boldsymbol{\sigma}$ close to memory $\boldsymbol{\xi}^1$. The activation going into neuron i will be

$$\sum_j W_{ij}\sigma_j = \frac{1}{N}\sum_j\sum_{\mu=1}^{M}\xi_i^\mu\xi_j^\mu\sigma_j \approx \xi_i^1 + \frac{1}{N}\sum_j\sum_{\mu=2}^{M}\xi_i^\mu\xi_j^\mu\sigma_j.$$

Our first step was to plug in the definition of W, and our second step was to separate out the $\mu = 1$ term as it is of special interest. The far right hand side is telling us that the Hopfield network wants to push neuron i to ξ_i^1—what it should be since $\boldsymbol{\sigma} \approx \boldsymbol{\xi}^1$—but there is also a noise term involving contributions from all the memories besides $\boldsymbol{\xi}^1$. In particular, this noise term is going to be of order M/N. Compare this to the Ising model, where every spin wants to be in the locally lowest energy orientation but is subject to thermal noise of order $k_{\mathrm{B}}T$. This suggests that M/N really is a sort of temperature for the Hopfield network.

Real ferromagnets demagnetize when heated, and an important prediction of the Ising model is that this demagnetization happens abruptly at a certain critical temperature. In greater than one spatial dimension, the Ising magnet has a Curie temperature where its magnetization craters as shown in the figure 5. The analogy we have developed in this section suggests that the same critical behavior we observe in the Ising model should also be observed in the Hopfield network. At some critical loading of memories, the net's retrieval should rapidly degrade just like the magnet's spins rapidly become disordered.

2.2.2 Numerically estimating the critical capacity

The Hopfield network's critical capacity has been found (see Amit et al. [1]) to be

$$M_{\text{critical}} = 0.138N. \tag{8}$$

We corroborate this finding by numerically simulating retrieval of corrupted memories in the classical Hopfield net. All the parameters mentioned here were chosen so that the simulation would take two to three hours to run. We set the number of neurons $N = 800$ and scan over the number of memories M from 2 to 151. At each M, we generate M random memories, use them to compute the weight matrix, and test the net's retrieval accuracy on 500 runs. On each run, we randomly sample a memory, corrupt 20% of the neuron activations within the memory by inverting them, and then test whether the net can converge back to the sampled memory within 100 steps. Collecting the results of all the runs, we are left with a series of retrieval accuracies over memory loadings, which we plot in the figure 6. The simulation code is available in a Jupyter notebook. Our results support a critical capacity somewhere in the range $0.070N$ to $0.150N$.

3 Problems and Generalization

3.1 Fused, Spurious Minima

Another limitation of the classical Hopfield networks, apart from its capacity limit, is its inability to distinguish between similar patterns. Hopfield noted in the 1982 paper that the network worked well if the stored patterns were randomly generated, so that any two patters are expected to have only 50% overlap with each other. However, when one reaches 80% or 90% overlap between stored patterns, Hopfield observed that the corresponding minima in the energy landscape often get fused together. When this happens, instead of retrieving a stored pattern, the network retrieves a superposition of the several stored patterns that make up the minima.

To reproduce this result, we downloaded some images from the internet and converted them to binary black and white pixel arrays, with black and white corresponding to ± 1. We stored these images in a Hopfield network of $128 \times 128 = 16,384$ neurons. According to the $0.138N$ capacity rule, this network should be able to store over $2,000$ patterns. However, our simulations show that even with 6 patterns, the situation described by Hopfield is already happening: minima corresponding to similar patterns are getting fused together, as shown in Figure 1.

3.2 Generalized Energy for Dense Associative Memories

To deal with the problem of fusing minima, one approach is to generalize the energy function to allow for sharper minima that can pull apart similar patterns. This approach is called Dense Associative Memory [8][7]. Instead of the square in the classical Hopfield energy function Eqn. (1), we allow for a general function F,

$$E(\boldsymbol{\sigma}) = -\frac{1}{2N} \sum_{\mu=1}^{M} F(\boldsymbol{\xi}^{\mu} \cdot \boldsymbol{\sigma}). \tag{9}$$

The corresponding update rule is derived in Section 3.3. An interesting case occurs when $F = \exp(\cdot)$, proposed in [4]:

$$E(\boldsymbol{\sigma}) = -\frac{1}{2N} \sum_{\mu=1}^{M} \exp(\beta \boldsymbol{\xi}^{\mu} \cdot \boldsymbol{\sigma}). \tag{10}$$

According to [13] and [2], after generalizing to continuous neurons, the update rule for minimizing energy in the Dense Associative Memory with this energy function becomes identical to the attention mechanism in [18].

We simulated a dense associative memory with energy function (10) and $\beta = 10$. It could indeed store up to 500 memories with high accuracy using just 100 neurons (see Figure 2), and it could recover the original image even from a very noisy starting pattern.

3.3 The Generalized Update Rule

To derive a rule for updating the state $\boldsymbol{\sigma}$, we can consider each neuron σ_i asynchronously (freezing the other $\sigma_{j \neq i}$) and determine which state of σ_i minimizes the energy by comparing $E(\boldsymbol{\sigma}, \sigma_i = +1)$ with $E(\boldsymbol{\sigma}, \sigma_i = -1)$. If $E(\boldsymbol{\sigma}, \sigma_i = +1) < E(\boldsymbol{\sigma}, \sigma_i = -1)$, we should set $\sigma_i = +1$ and if $E(\boldsymbol{\sigma}, \sigma_i = -1) < E(\boldsymbol{\sigma}, \sigma_i = +1)$, we should set $\sigma_i = -1$ because an energy minimum corresponds

to a stored memory. Explicitly writing out the energy formulas for $E_{+1} = E(\boldsymbol{\sigma}, \sigma_i = +1)$ and $E_{-1} = E(\boldsymbol{\sigma}, \sigma_i = -1)$ we get:

$$E_{+1} = -\sum_{\mu=1}^{K} F\left[\sum_{j\neq i} \xi_j^\mu \sigma_j + \xi_i^\mu\right] \tag{11}$$

$$E_{-1} = -\sum_{\mu=1}^{K} F\left[\sum_{j\neq i} \xi_j^\mu \sigma_j - \xi_i^\mu\right] \tag{12}$$

Consider the case $E_{-1} < E_{+1}$, so $E_{-1} - E_{+1} < 0$. The state $\boldsymbol{\sigma}$ with $\sigma_i = -1$ is preferred. Symmetrically, for $E_{+1} < E_{-1}$ (meaning $E_{-1} - E_{+1} > 0$) the system should set $\sigma_i = +1$ to minimize its energy. We can formulate this idea into the following update rule:

$$\boldsymbol{\sigma}_i^{t+1} = sign\left(E_{-1} - E_{+1}\right) \tag{13}$$

Which we can expand and manipulate to yield the update rule from [8]:

$$\boldsymbol{\sigma}_i^{t+1} = sign\left(-\sum_{\mu=1}^{K} F\left[\sum_{j\neq i} \xi_j^\mu \sigma_j - \xi_i^\mu\right] - \left(-\sum_{\mu=1}^{K} F\left[\sum_{j\neq i} \xi_j^\mu \sigma_j + \xi_i^\mu\right]\right)\right) \tag{14}$$

$$\boldsymbol{\sigma}_i^{t+1} = sign\left[\sum_{\mu=1}^{K}\left(F\left(\xi_i^\mu + \sum_{j\neq i} \xi_j^\mu \sigma_j\right) - F\left(-\xi_i^\mu + \sum_{j\neq i} \xi_j^\mu \sigma_j\right)\right)\right] \tag{15}$$

3.3.1 Practical demonstration

For purposes of demonstration, a publicly available web tool was made available.[5] This playground enables a user to experience the issue of crosstalk and spurious fused states through interaction with a drawable pixel grid. The user draws a character, then uses the original Hopfield associative memory update rule to converge to a pre-embedded character. When seven correlated memories are packed into the system instead of just three, the user may experience an issue like the one in Figure 4. The issue is resolved through the use of the Dense Associative Memory update rule which is present on another grid on the right side of the screen.

4 Biological Implications

4.1 Application to an LIF neuron network

The discussion in the previous section begs this question: if dense associative memories are so great, why are classical Hopfield networks still valuable? The reason is biological plausibility. The weight matrix in the classical network has the direct biological interpretation as synaptic connection

[5] Demo: https://drawfield.vercel.app/ - Code: github.com/ssocolow/drawfield.

strengths between neurons, so we can easily incorporate the Hopfield concept into concrete biology-based models like leaky integrate-and-fire, where W_{ij} just tells us how strongly an action potential from neuron j affects the membrane potential of neuron i. Mathematically, we use the equations

$$u_i(t) = \sum_f \eta_i(t - t_i^f) + \alpha \sum_j W_{ij} \sum_f \varepsilon_{ij}(t - t_j^f) + \text{escape noise},$$

$u_i(t)$ is the membrane potential of neuron i, $\eta_i(t - t_i^f)$ describes the refractoriness[6] of a neuron after it has generated a spike at time t_i^f, $\varepsilon_j t(t - t_j^f)$ describes the effect on neuron i of a spike generated by neuron j at time t_j^f, W_{ij} are the Hopfield weights corresponding to the stored patterns, α is an empirically tuned parameter to make the model work with Hopfield, and the escape noise is added to model background stochastic firing along the lines of Eqns. (9.2) and (9.3) in [5]. η and ε are set according to Equations (6.56) in [5].

We simulated the dynamics of the $u_i(t)$ for a network of 30 neurons storing 3 patterns. Figure 3 shows how membrane potentials of all 30 neurons vary over time; it has indeed succeeded in retrieving one of the stored patterns, as indicated by the persistently firing red neurons.

However, the energy functions for dense associative memories are cubic or higher order, so the weight matrix becomes a higher order tensor and the biological interpretation disappears. Krotov and Hopfield [9] suggest that we could explain higher order interactions in terms of hidden layers of neurons with simple synaptic connections, so this could be a plausible future research direction.

4.2 Asymmetric Weight Matrices

In his 1982 paper [6], Hopfield acknowledged that real neurons need not have symmetric connections, so he also considered the possibility of asymmetric weight matrices. Using simulations with randomly generated asymmetric $\mathbf{W}$, Hopfield exhibited the existence of fixed points, cycles, or chaotic wandering, in a fashion similar to the phase space dynamics examined in this course. He concluded that both symmetric and asymmetric weight matrices have "the properties necessary for a physical content-addressable memory". To explain why attractors exist even for asymetric $\mathbf{W}$, Hopfield pursued a line of reasoning similar to what our group did in Section 2.2. Suppose we have an asymmetric weight matrix $\mathbf{W}$ and at some step, σ_i is flipped to $\sigma_i' = \sigma_i + \Delta\sigma_i$ according to the update rule Eqn. (4). Then we have

$$\Delta E = E' - E = -\frac{1}{2}\sum_{k\neq l} W_{kl}\sigma_k'\sigma_l' - \left(-\frac{1}{2}\sum_{i\neq j} W_{kl}\sigma_k\sigma_l\right) = -\frac{1}{2}\sum_{k,l} W_{kl}\left(\sigma_k'\sigma_l' - \sigma_k\sigma_l\right).$$

Since only σ_i has been flipped, we only need to consider the terms where k or l is equal to i:

$$\Delta E = -\frac{1}{2}\sum_{k,l} W_{kl}\left(\sigma_k'\sigma_l' - \sigma_k\sigma_l\right) = -\frac{1}{2}\sum_{k} W_{ki}\left(\sigma_k'\sigma_i' - \sigma_k\sigma_i\right) - \frac{1}{2}\sum_{l} W_{il}\left(\sigma_i'\sigma_l' - \sigma_i\sigma_l\right).$$

This expression is not fully correct, since the term $W_{ii}\left(\sigma_i'\sigma_i' - \sigma_i\sigma_i\right)$ is considered in both sums. However, since this term is zero anyway, we can just explicitly exclude it from the summation. Observing that $\sigma_k' = \sigma_k$ whenever $k \neq i$, we have

$$\Delta E = -\frac{1}{2}\sum_{k\neq i} W_{ki}\sigma_k\Delta\sigma_i - \frac{1}{2}\sum_{l\neq i} W_{il}\sigma_l\Delta\sigma_i = -\frac{1}{2}\Delta\sigma_i\left(\sum_{k\neq i} W_{ki}\sigma_k + \sum_{l\neq i} W_{il}\sigma_l\right)$$

[6] The use of η eliminates the need for an explicit reset clause in the Euler method code.

If $\mathbf{W}$ were symmetric, the two sums would be identical, and we would have

$$\Delta E = -\Delta\sigma_i \sum_{k\neq i} W_{il}\sigma_l,$$

reproducing the energy formula Eqn. (3). When $\mathbf{W}$ is asymmetric, however, the term

$$-\frac{1}{2}\Delta\sigma_i \sum_{l\neq i} W_{il}\sigma_l$$

is still manifestly negative by the update rule for σ_i, while the "stochastic" term

$$-\frac{1}{2}\Delta\sigma_i \sum_{k\neq i} W_{ki}\sigma_k$$

could be either positive or negative. When the weights in $\mathbf{W}$ are chosen randomly, the sum $\sum_{k\neq i} W_{ki}\sigma_k$ fluctuates around zero. This is why Hopfield states that the energy trajectory for asymmetric $\mathbf{W}$ is analogous to the trajectory of a symmetric $\mathbf{W}$ (since there is a stable, negative contribution to ΔE, driving the system towards an energy minimum), but with a finite temperature (corresponding to the indeterminate stochastic term).

Asymmetric Hopfield weights could lead to some interesting dynamics. For example, Hopfield notes that given a specific modification to the symmetric $\mathbf{W}$,[7] a system has been observed to spend a while near one stored state, then leave to go to a point near another stored state, thereby exhibiting a cascade through several metastable states in the energy landscape. However, Hopfield did not elaborate on the biology behind the dynamics of asymmetric networks. We explore this topic in the Section 4.3 below, using more recent literature.

4.3 A Survey of Relevant Neuro-biological Literature

In 2021, Hopfield reiterated the biological plausibility of the classical model by noting the lack of coherence in dense associative networks (as seen in 4.1) [9]. Consider a three-neuron model, for example: the energy function describing a classical Hopfield network is

$$E = -\sum_{i,j} T_{i,j}\sigma_i\sigma_j$$

while for a dense associative network we have

$$E = -\sum_{i,j,k} T_{i,j,k}\sigma_i\sigma_j\sigma_k$$

where the product of two neurons σ_i, $\sigma_j \in \{\sigma_1, \sigma_2, \sigma_3\}$ represents a synaptic connection. It follows that the addition of a third neuron, though seemingly innocuous, breaks with neuronal physiology. In this dense associative network, a "sending" neuron might deliver neurotransmitter signals to two "receiving" neurons via the *same* axon branch; neurons in the brain, by contrast, interface uniquely such that there is one axon branch for each "receiving" neuron. It is certainly possible to have a connection of three neurons, but within each individual synapse only two distinct neurons may connect.

This motivating example is relevant in that it speaks to another underlying problem with Hopfield networks: what does it mean to be biological?

[7] This modification is defined in Equation 13 in [6].

Trivially, one might define biological plausibility in that the mathematical techniques used in modeling match or mirror localized phenomena (such as classical Hopfield networks realizing a biological synapse) that occur in neurophysiology. There is merit to this description, especially in understanding that models of such complex behavior rarely account for every possible dynamic. Still, it seems reasonable to ask "if Hopfield networks are biologically realizable, are they empirically observed?"

They are! In 2014, computational biologists argued that the pattern signatures in the CA3 reigon of the hippocampus follow those exhibited by Hopfield networks, specifically *asymmetric* networks (see section 4.2) [14]. Indeed, it is not just a fun tangent to show that asymmetric networks have analogous energy functions, as they are pivotal to neuro-biological models. When an input is given to an asymmetric network, the directionality of weights will enable the transfer from fixed-pattern to fixed-pattern. This leads to a cascading of imagery, where one stored pattern is accessed, is transfigured into the next, and so on. This "strong evidence" for neuronal behavior first established in 2014 has since accumulated and morphed into a wealth of biological applications with rich feature-sets. Below is a cursory overview of some of this literature:

4.3.1 A Generative Model of Memory Construction and Consolidation

Spens and Burgess' work of the same title sets up a powerful framework for a more distributed description of memory [16]. They implicate the hippocampus, as above, but add in the neo-cortex. This is the outer region of the brain, typically associated with higher order functioning (and in this case long-term memory consolidation). Memory, then, is as follows: an event, depicted in sequences of imagery like the patterns in a Hopfield network is stored in "one shot" (i.e. stored promptly after it occurs). The stored memory landscape, then, plays the role of the teacher, helping to train the neo-cortex as a generative neural-net model (the student). This reinforcement learning schema is sufficient to depict episodic memory and even imagination, as the generative model can perform detailed recall with sensory input. Interestingly, the authors opted to deploy a dense associative network for the hippocampus, finding realism at the level of systems neurophysiology rather than at the cellular level. Nonetheless, combined with the next paper, the use of Hopfield networks is salient in that it can also provide an account of dreaming.

4.3.2 Predictive Sequence Learning in the Hippocampal Formation

Chen and colleagues refine the original research on Hopfield associative function in the hippocampus by defining the role of the CA1 region relative to the CA3 [3]. Where the CA3 performs classical pattern recall in the form of predictive visual sequences (from sensory input), the CA1 places the predicted sequence against the next available external stimuli to compute error and refine CA3 output. Meaningfully, this is a biological process under which patterns with significant overlap that lead to spurious minima can be addressed twofold: changes to the CA3 network due to the error-correcting role of the CA1 help to create a more accurate predictive network, and the actual patterns that are then stored are de-correlated (often referred to as "whitening"). When the system is no longer taking in external stimuli, for example during sleep, the CA3 region can follow the asymmetric landscape and move sequentially through these accurate and newly de-correlated episodic memories. Put together with the paper above, we see that the hippocampal replay of episodic memories in sleep is training the neocortex for long-term memory. Indeed, dreams might be explained by the dynamics of Hopfield networks!

4.3.3 Sequential Memory with Temporal Predictive Coding

This last paper delves into the world of linguistic (rather than visual) memory, still using asymmetric Hopfield networks (AHNs) to recall sequences of patterns in the form of words. Authors Tang et al. employ "temporal predictive coding (tPC)", a family of models using similar logic to 4.3.2 that acts on words instead of images [17]. They present two versions, one a classical asymmetric Hopfield network and the other a two-layer tPC that uses a hidden layer to transform external stimuli. Critically, this paper explores some key topics in Hopfield networks: For one, they pit this single-layer tPC with asymmetric weights against the dense associative network model. With the whitening process described in 4.3.2, the single-layer tPC *outperforms* the dense network in accuracy and sequential recall tasks, though the exponential memory of the dense network allows for more pattern storage relative to the number of neurons. Because of the accuracy, however, tPCs are important in modeling human behavior. The authors employed the 2-layer tPC in a word-sequence recall task (a measure of memory span) and were able to more convincingly match human performance shown by Psychologists Crannel and Parish's seminal work from 1957 as compared to a 2006 recurrent neural network model. It is plausible once again that Hopfield networks might be the most appropriate model of explaining how these dynamics work in the brain.

With some certainty we can argue that Hopfield networks are not just a theoretical framework for understanding neurobiological processes, but might be exactly how the brain performs cognition both at the cellular and the systems level. It's only 43 years old- what's next for these networks?

5 Breakdown of Work

5.1 Simon

Explained the dense associative memory update rule and intuition in presentation and derived it in paper. Wrote web app implementing stochastic synchronous original Hopfield updates and asynchronous dense associative memory updates. Demonstrated app in presentation and described it in paper. Code in the file `drawfield-main.zip`

5.2 Miles

Wrote section 2.2 on the critical capacity and the Hopfield-Ising analogy. Programmed and ran simulation in the file `HopCapacity.ipynb` to rederive the critical capacity numerically. Explained the Ising model and its relevance to Hopfield nets during the presentation.

5.3 Neel

Expounded on the biological trouble with dense associative memory energy functions, introducing literature that identifies real-life examples of Hopfield Networks in humans. Contributed the idea of asymmetric networks (though the math was subsequently explained by Forrest in 4.2) as a means to novel computational models of Hopfield networks in the brain. Presented on biological plausibility, asymmetric weights, their corresponding energy landscape, and biologically-related computational models using Hopfield networks to achieve various facets of memory (and dreaming).

5.4 Forrest

Wrote Sections 1.2, 1.3, 2.1, 3.1, 3.2, 4.1, and 4.2 in the paper. Wrote Hopfield model, Dense Associative Memory, and Hopfield-LIF network implementations in the file `Final Project Hopfield.ipynb`. Talked about the classical network, its problems, Dense Associative Memories, and the Hopfield-LIF network during presentation.

References

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6 Figures

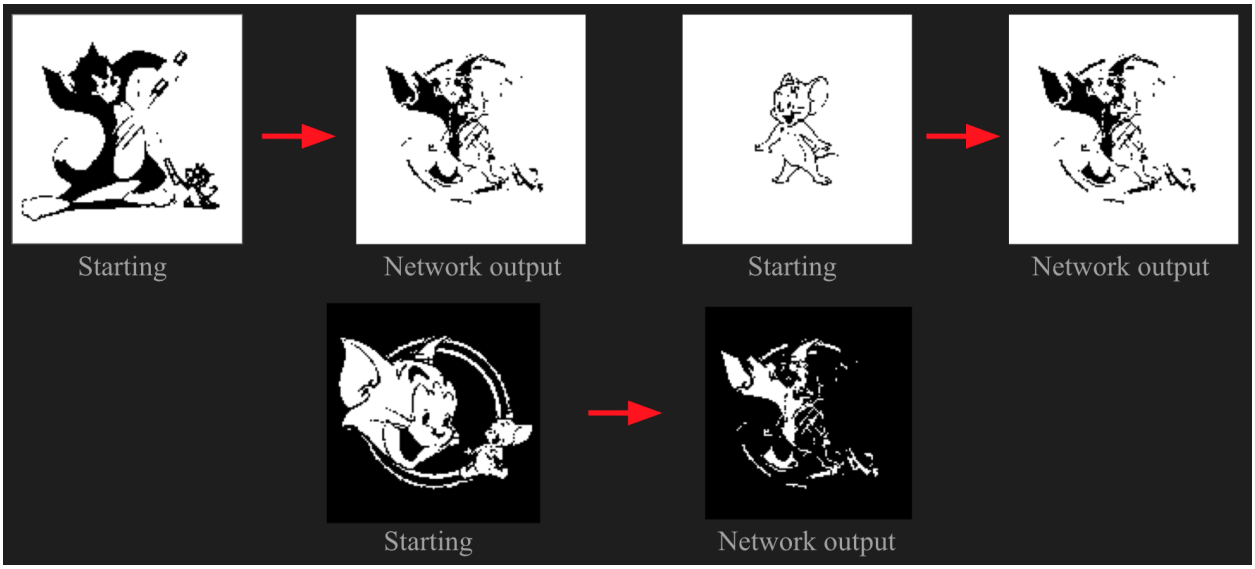

Figure 1: Fused minima corresponding to three similar patterns. Note that the patterns here are "similar" because they are all predominantly black or predominantly white. Since the energy function is a sum of *squares* of dot products, it does not care about the actual sign of the dot product, but only its absolute magnitude. Thus a completely black and a completely white image would have the highest possible overlap, just like two black ones or two white ones.

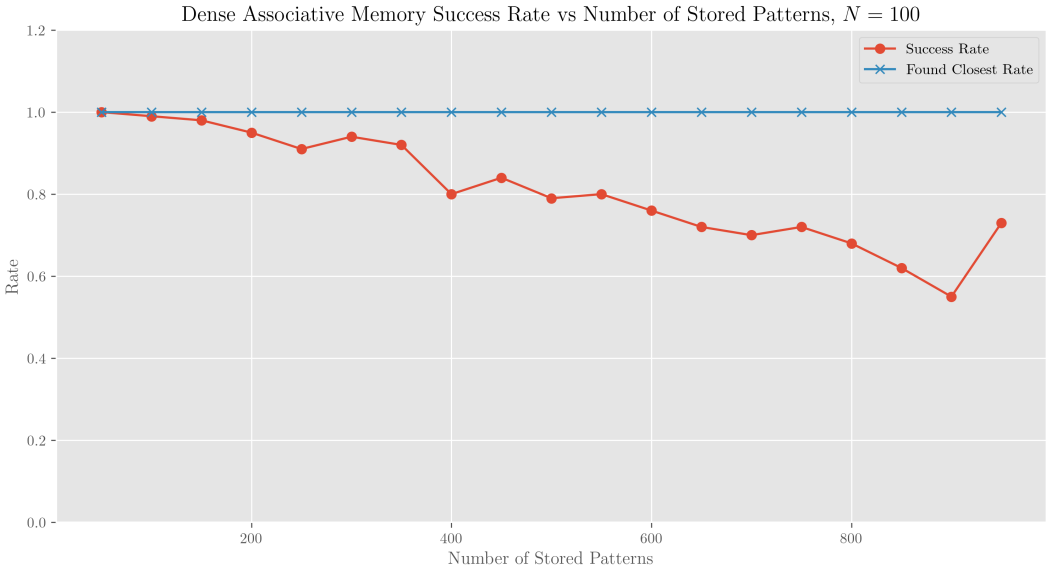

Figure 2: Successful retrieval rate vs. number of stored patterns of a 100-Neuron dense associative memory with exponential energy function. We choose the starting pattern by selecting one stored pattern and randomly flipping 30% of the values in its bit array. The "Success Rate" is the probability that the network recovers the stored pattern from which the starting pattern was modified. The "Found Closest Rate" is the probability that the starting pattern eventually converges to the stored pattern that is closest to it, regardless of which pattern the starting pattern was actually modified from. We see that found closest rate remains 100%, so all errors in retrieval are due to the starting pattern becoming closer to a different pattern than its parent pattern after the bit flip process.

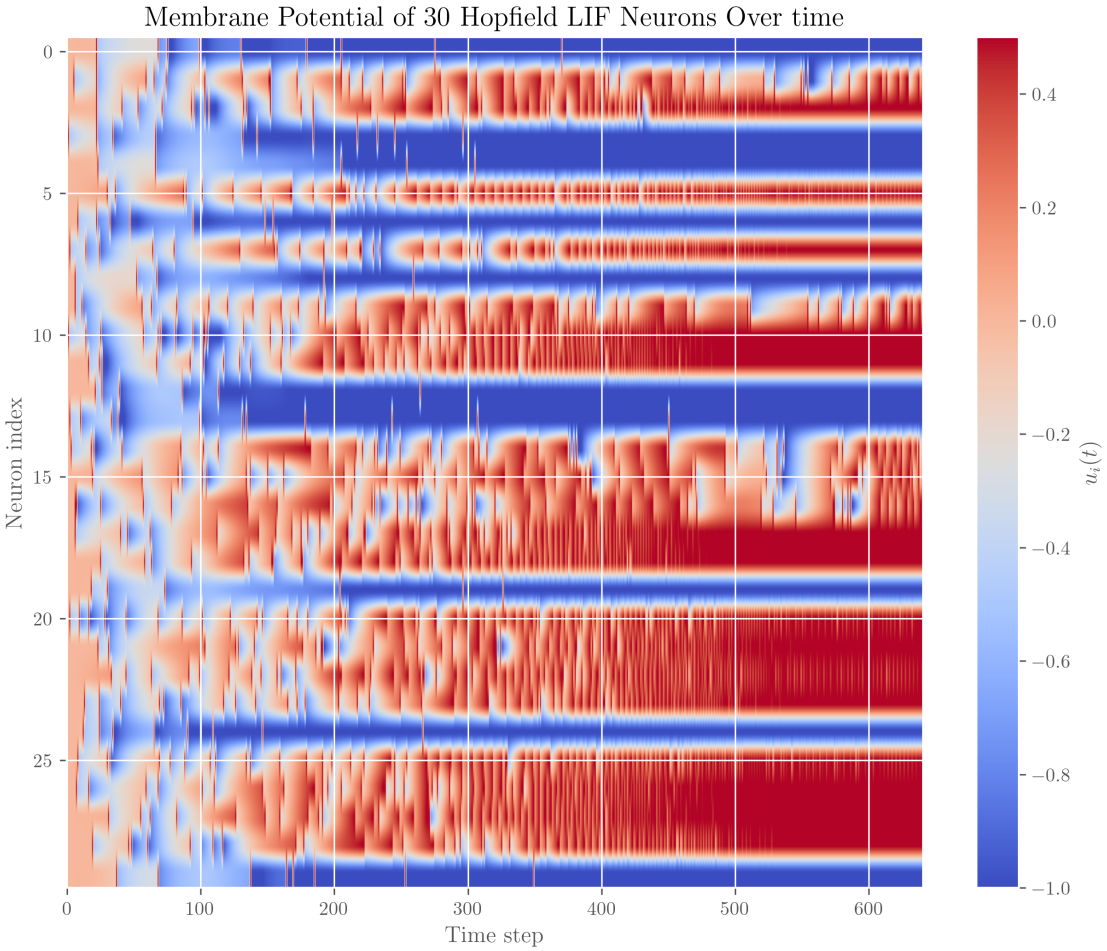

Figure 3: A Hopfield-LIF Network retrieving a pattern. Note how the firing gradually builds up over time, with uncertain sporadic firing followed by a much more resolute pattern.

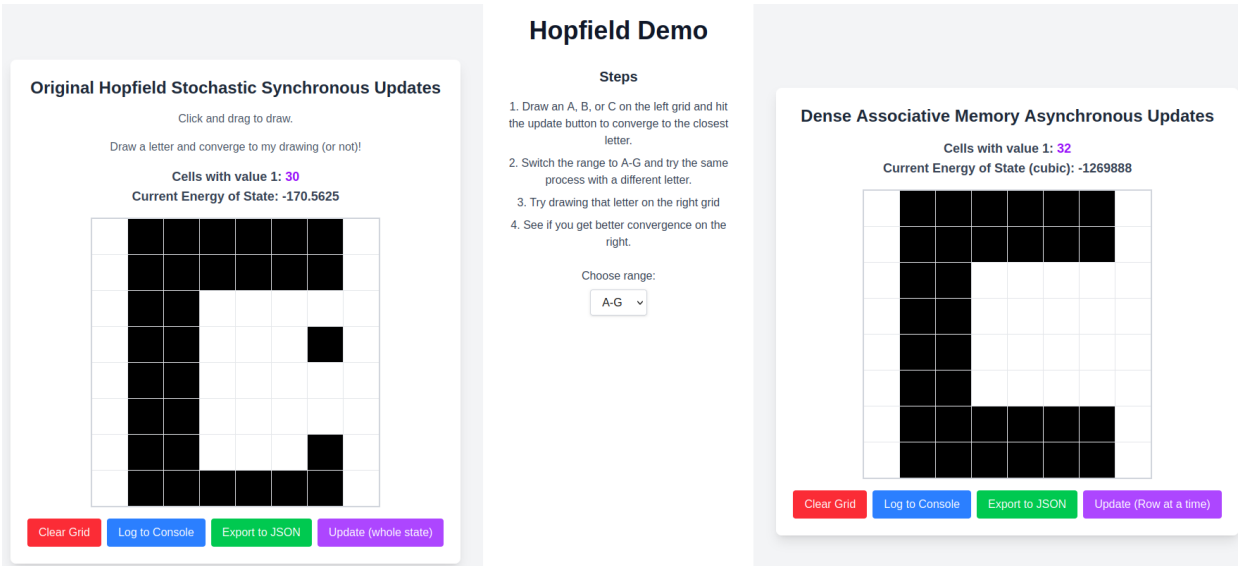

Figure 4: Web demonstration showing the original Hopfield update rule (left) converging to a spurious fused state (between a C and a G) while a cubic energy function used in the generalized Dense Associative Memory update rule (right) converges without issue.

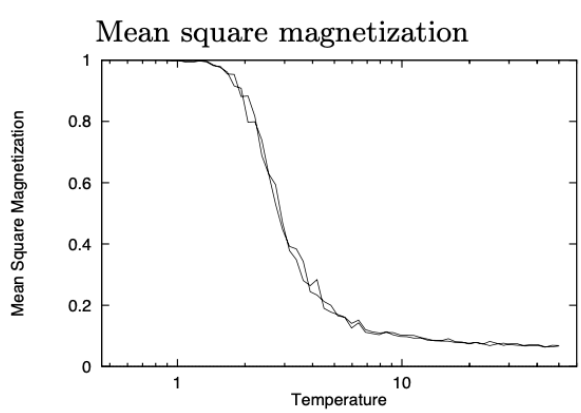

Figure 5: Criticality in the Ising magnet. This plot shows results from monte carlo simulations of the Ising model over a range of temperatures. The mean square magnetization (vertical axis) measures how aligned or "ordered" the spins are on average. Figure courtesy of [10]

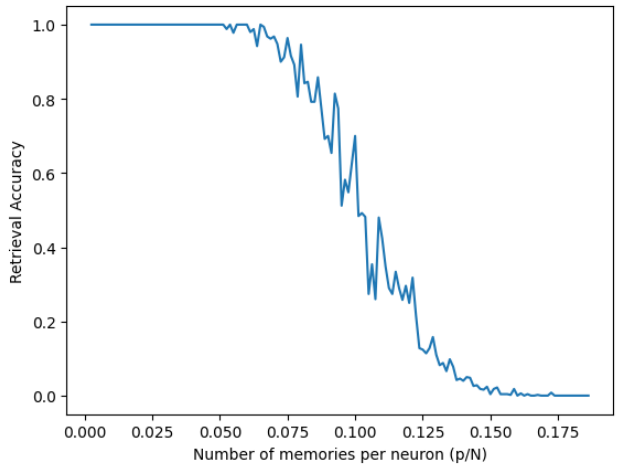

Figure 6: Estimating the critical capacity from scratch. This plot was generated by us using the procedure explained in section 2.2.